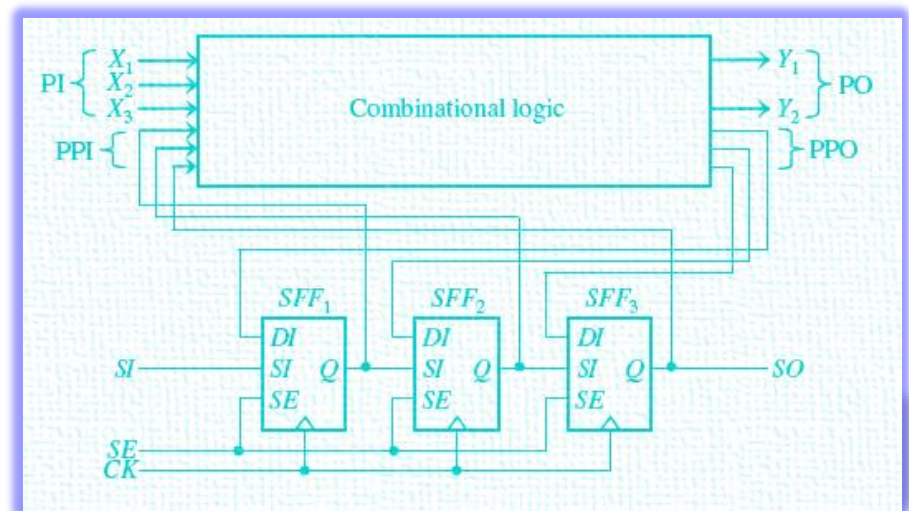
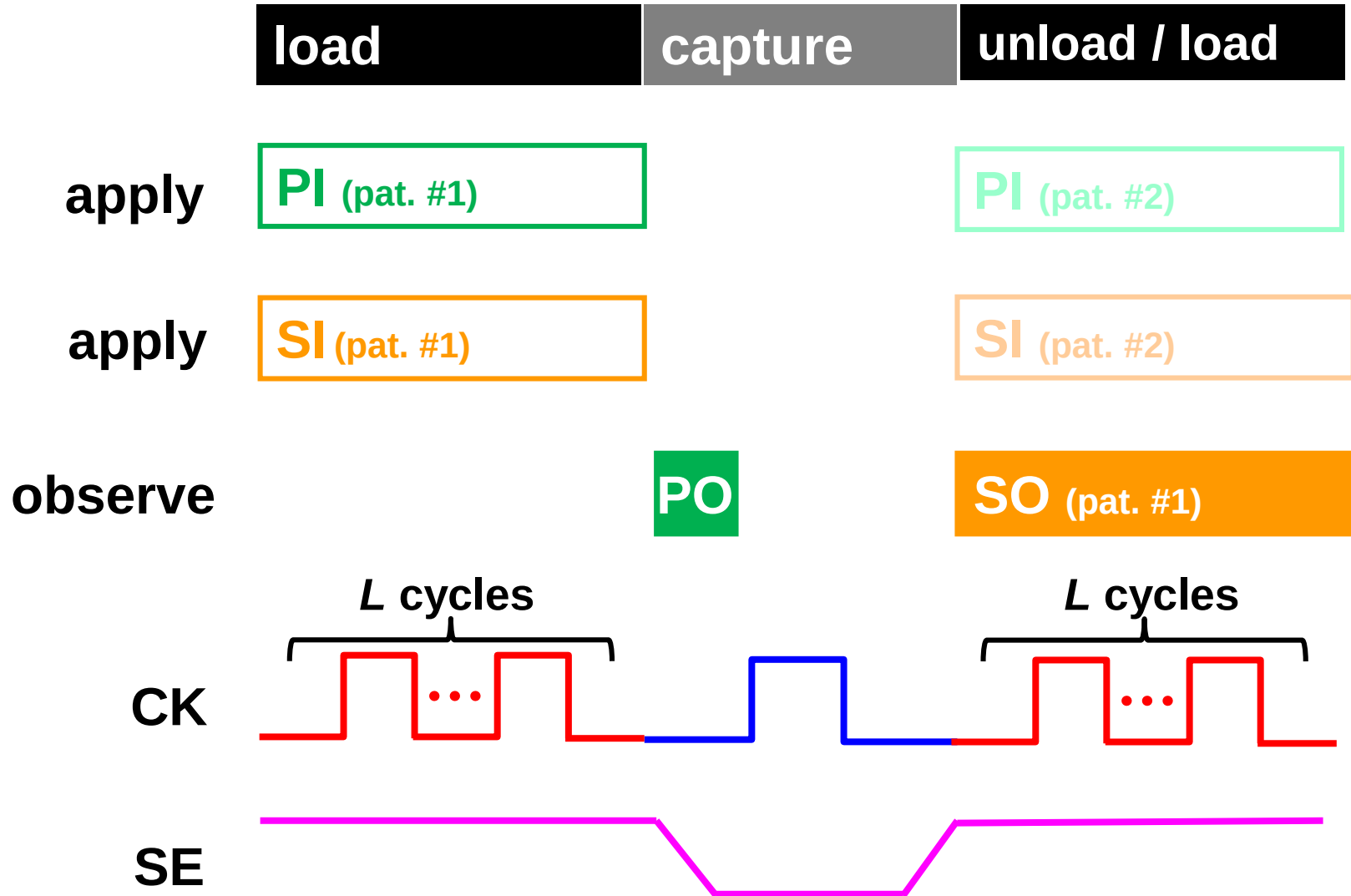


DFT - Part 1

- Introduction
- Internal Scan
- Scan Design Flow
- Issues and Solutions
 - ◆ Long test time
 - ◆ Large test data
 - ◆ Too much overhead
 - ◆ How to Test DFT
- Conclusion



SSF Operation (review)

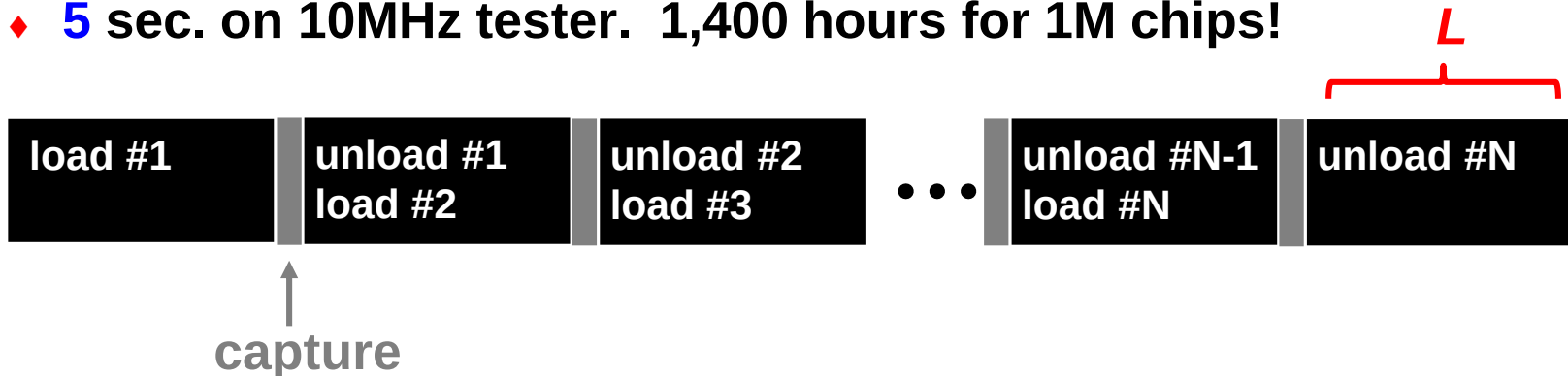


Issue #1: Long Test Time

- How many test cycles do we need?
 - ♦ L = length of scan chain ; $N_{pattern}$ = number of test patterns

$$\underbrace{(N_{pattern} + 1) \times L}_{\text{shift}} + \underbrace{N_{pattern}}_{\text{capture}}$$

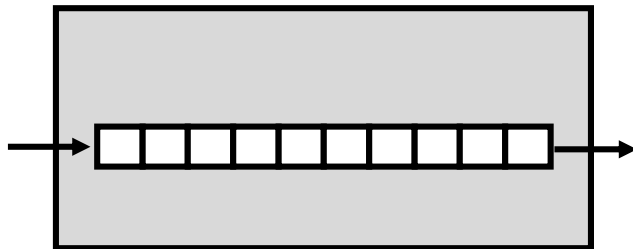
- Example: Apply 5,000 test patterns to a CUT of 10K SFF
 - ♦ Single scan chain $L = 10,000$; $N_{pattern} = 5,000$
 - ♦ Total **50,015,000** cycles
 - * $(5,000+1) \times 10,000$ shift cycles
 - * 5,000 capture cycles
 - ♦ **5** sec. on 10MHz tester. 1,400 hours for 1M chips!



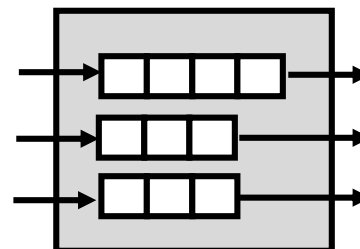
How to Save Test Time?

- To shorten L , we can partition SFF into n scan chains
 - ♦ L is determined by *longest* scan chain in CUT
 - * because ATE shifts all scan chains together
 - ♦ Penalty: More scan I/O pins (see FFT)
- Example: 10K SFF partitioned into three scan chains
 - ♦ $L = \max \{ 3K, 3K, 4K \} = 4K$
 - ♦ 60% reduction in test time!

Single scan chain
 $n=1, N_{SFF}=10K, L=10K$



Three scan chains
 $n=3, N_{SFF}=1K, L=4K$



Multiple Chains Reduce Test Time

Quiz

Q1: Apply 10K patterns to CUT of 6M SFF in three scan chains

$$L_1 = 1M ; L_2 = 1M ; L_3 = 4M$$

How many cycles do we need to test this CUT?

Q2: What if we balance scan length: $L_1 = 2M ; L_2 = 2M ; L_3 = 2M$

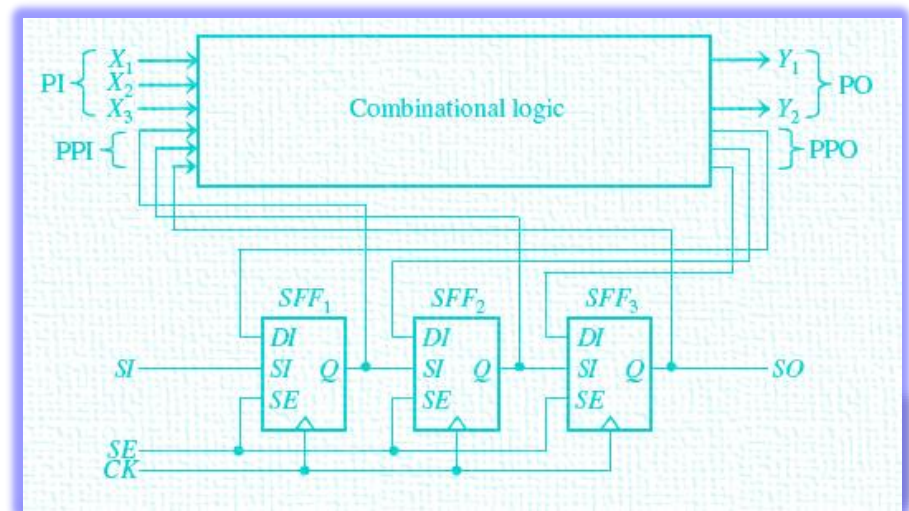
$$\underbrace{(N_{pattern} + 1) \times L}_{\text{shift}} + \underbrace{N_{pattern}}_{\text{capture}}$$

A1:

A2:

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Issue #2: Large Test Data

- How many test data stored on ATE?

Scan in data: $N_{pattern} \times N_{SFF}$

Scan out data: $N_{pattern} \times N_{SFF}$

PI data: $N_{pattern} \times N_{PI}$

PO data: $N_{pattern} \times N_{PO}$

N_{SFF} = number of scan FF

N_{PI} = number of PO

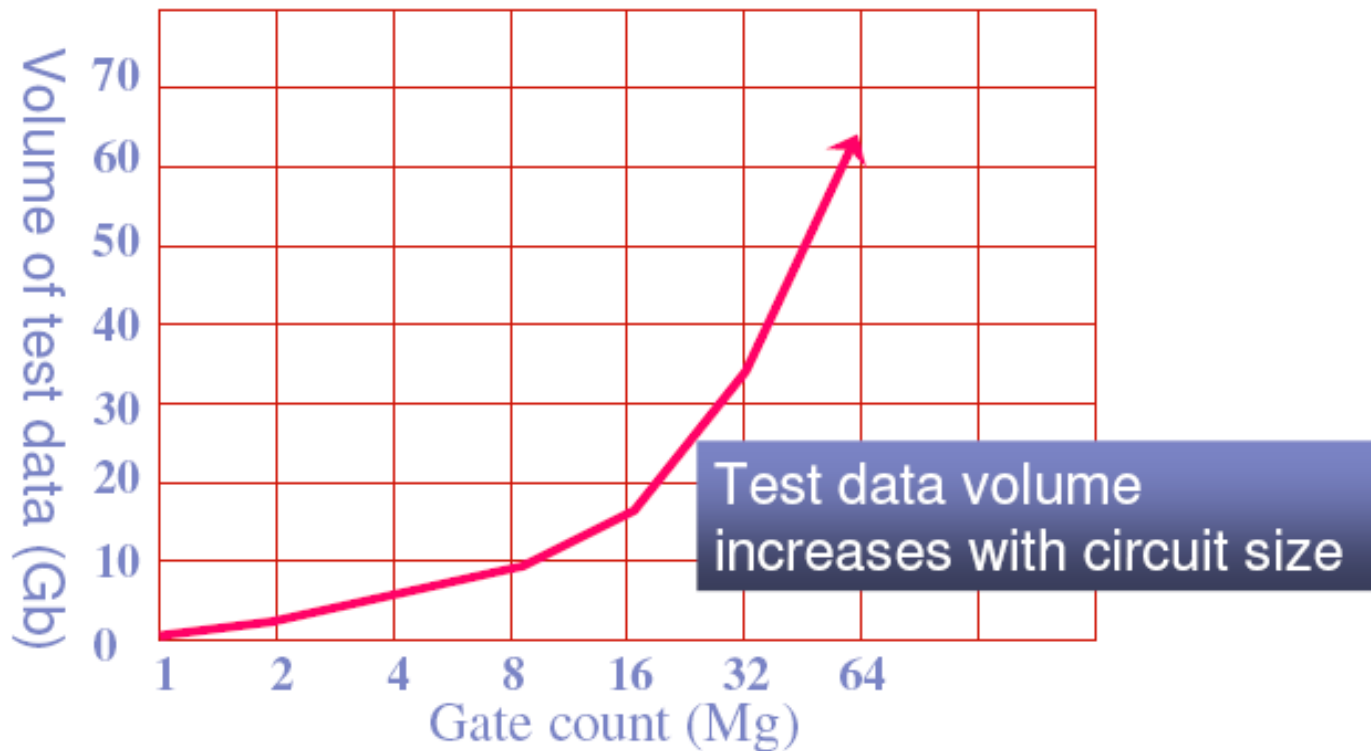
N_{PO} = number of PI

- Example:

- ♦ Apply 10K patterns to CUT of 6M SFF
- ♦ Suppose $N_{PO}=50$, $N_{PI}=20$
 - * PI data = 10K x 20 = **200M bits**
 - * PO data = 10K x 50 = **500M bits**
 - * Scan in data = 10K x 6M = **60G bits!**
 - * Scan out data = 10K x 6M = **60G bits!**

Test Data Volume Skyrockets

- Suppose ATE has 500 pins, each has 64Mb memory
 - ♦ Total ATE memory available = **32Gb**
 - ♦ 60Gb cannot fit in ATE!
- Q: Does multiple scan chain reduce test data volume?



(Source: Blyler, Wireless System Design, 2001)

Quiz

Q: Apply 10K patterns to CUT of 6M SFF in **three scan chains**

$L_1 = 2M$; $L_2 = 2M$; $L_3 = 2M$

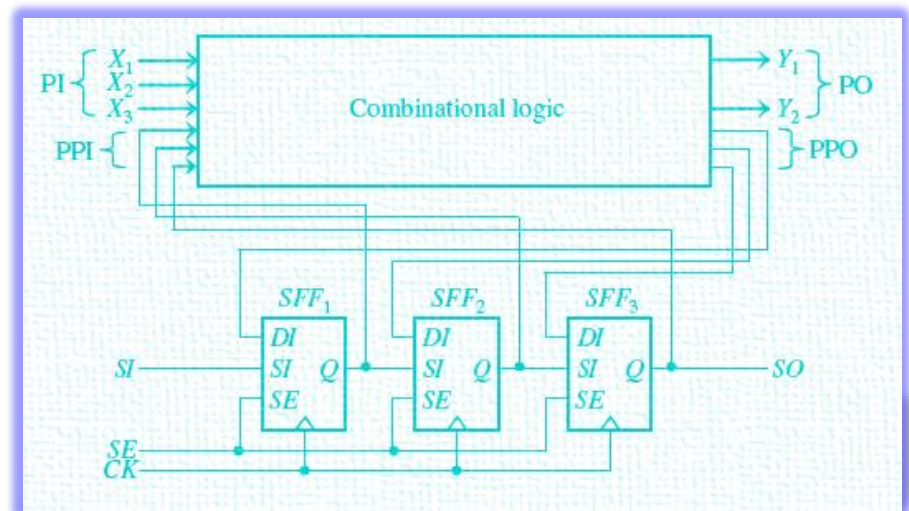
How many scan test data do we need?

A:



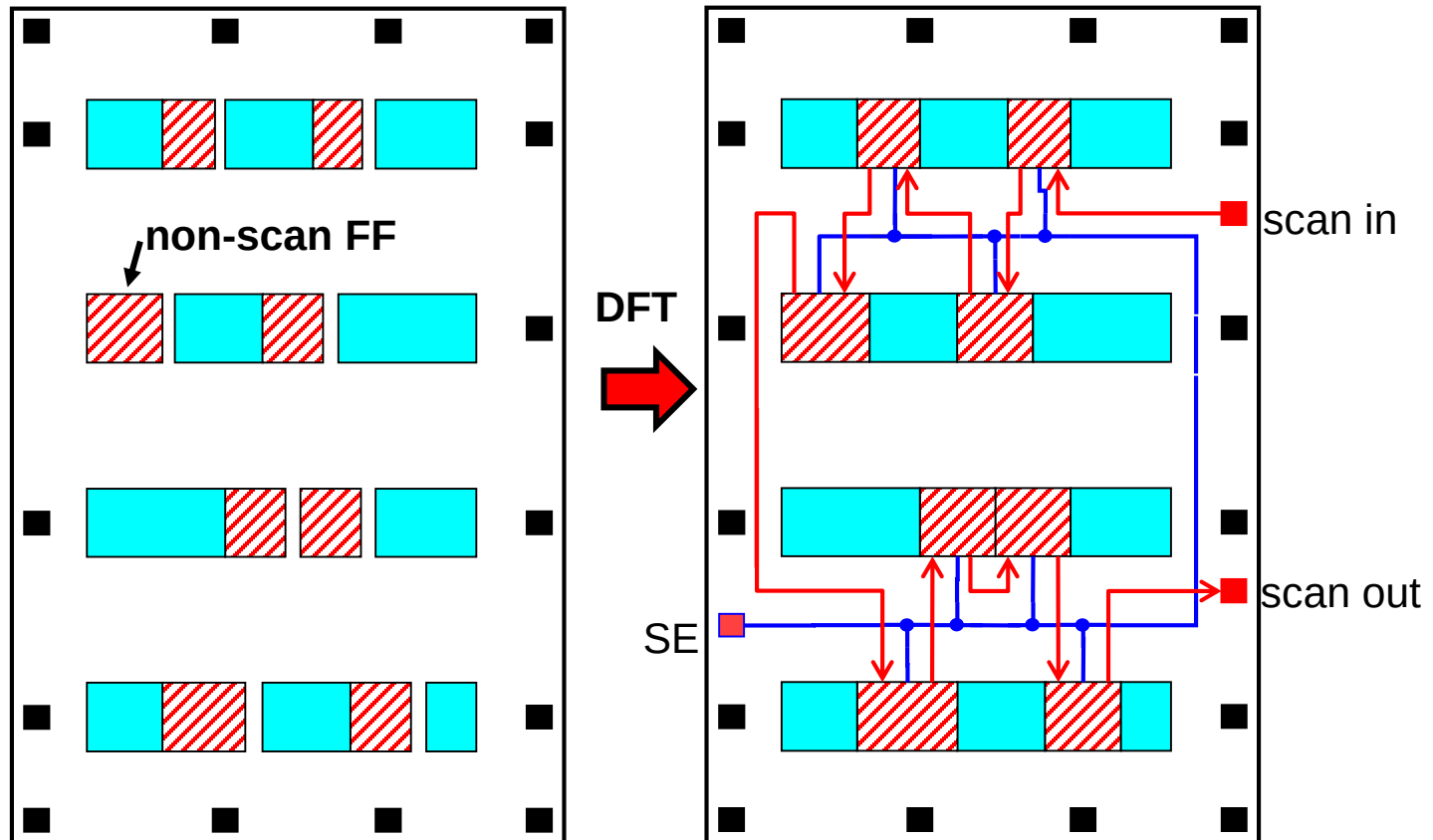
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Issue #3: Too Much Overhead

- Area overhead, Timing overhead, Power overhead

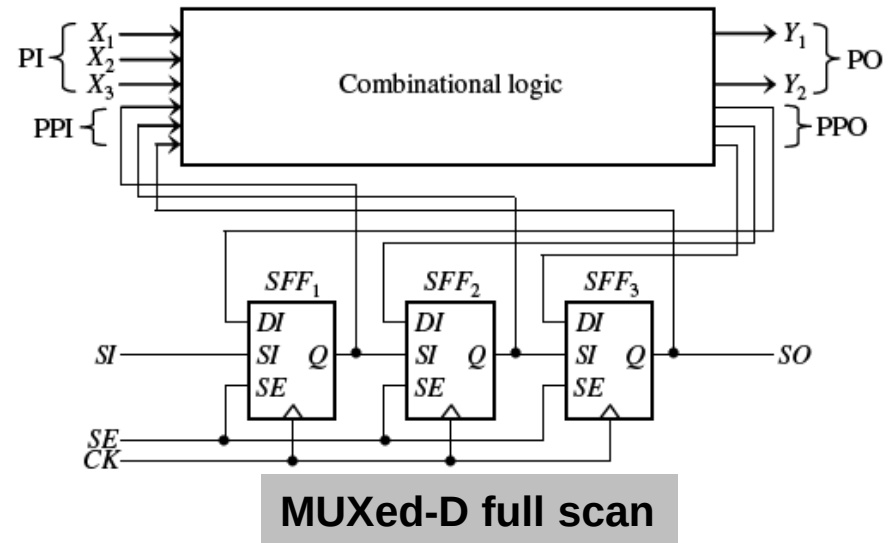


DFT about 5% Timing, 10% Area/Power Overhead

How to Reduce Overhead? Partial Scan

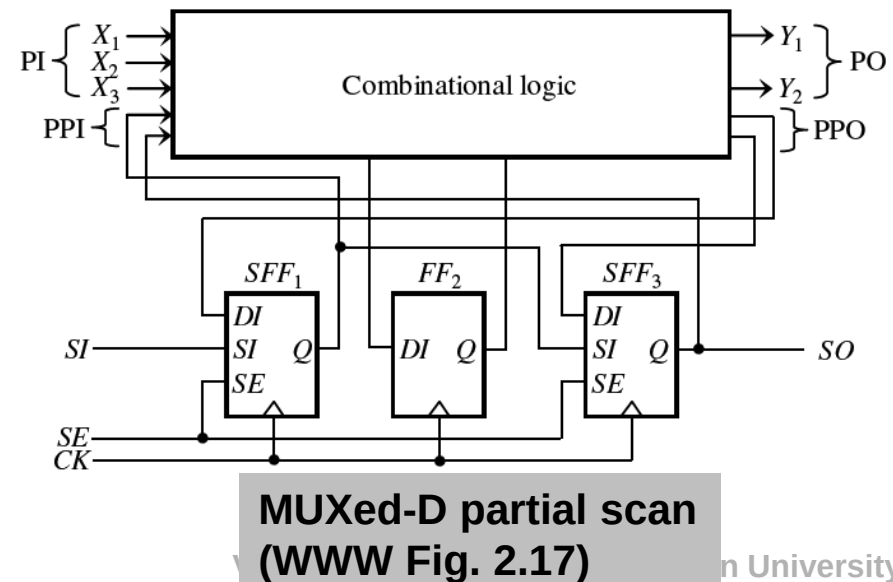
- **Full Scan**

- ♦ Every FF scannable
- ☹ More overhead
- 😊 Higher fault coverage
- 😊 Shorter ATPG run time

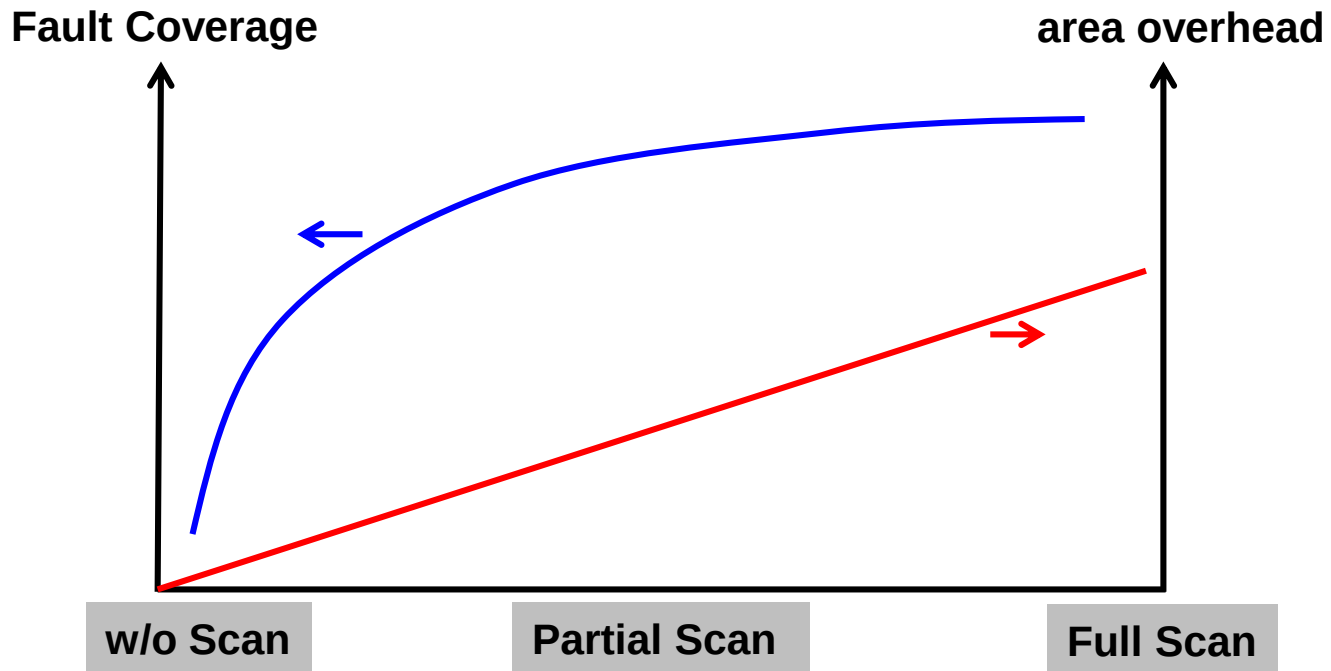


- **Partial Scan**

- ♦ Some FF not scannable
- 😊 Less overhead
- ☹ Lower fault coverage
- ☹ Longer ATPG run time



Trade-off Area and FC



S5378 Example (BA Table 14.1 modified)

	w/o scan	Partial scan	Full scan
no. of logic gates (N_L)	2,781	2,781	2,781
no. of non-Scan FF (N_{NFF})	179	149	0
no. of Scan FF (N_{SFF})	0	30	179
area overhead*	0.0%	1.63%	9.73%
no. of faults	4,603	4,603	4,603
(PI+PPI) / (PO+PPO)	35/49	65/79	213/228
Fault Coverage	70.9%	93.7%	99.1%
CPU time (SUN 200MHz)	5,533 s	727 s	5 s
no. of Test Patterns	414	1,117	585
no. of Test Cycles	414	34,657	105,479

$$*overhead = \frac{Scan - w/o_scan}{w/o_scan}$$

1 non-scan FF = 5 logic gates
1 scan FF = 7 logic gates

Quiz

Q: Please calculate partial scan area overhead

Assume 1 non-scan FF = 5 logic gates

Assume 1 scan FF = 7 logic gates

$$*overhead = \frac{Scan - w/o_scan}{w/o_scan}$$

	w/o scan	Partial scan	Full scan
no. of logic gates (N_L)	2,781	2,781	2,781
no. of non-Scan FF (N_{NFF})	179	79	0
no. of Scan FF (N_{SFF})	0	100	179
area overhead	0.0%	?	9.73%

A:

CH1: DFT or Not?

- Q: Is it economical to insert DFT?
- A: Yes. This is true for many products.

$$DL = 1 - Y^{(1-FC)}$$

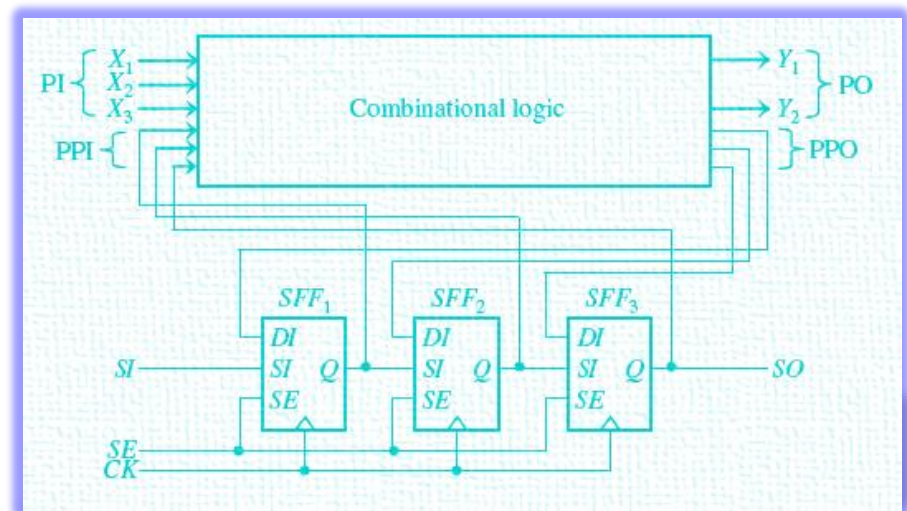
- ♦ Although Y drops, DL improves significantly

Item	w/o DFT	with DFT
Total # of Dies	1,000,000	900,000
Yield	98%	97%
FC fault coverage	70%	99%
$DL = 1 - Y^{(1-FC)}$	6,043 DPM	304 DPM
Sales = $D \times Y \times \$1$	980,000	873,000
Repair cost = $D \times Y \times DL \times \100	592,163	26,587
Profit = $S - R$	387,837	846,413

Despite Overhead, DFT Is Worth Doing!

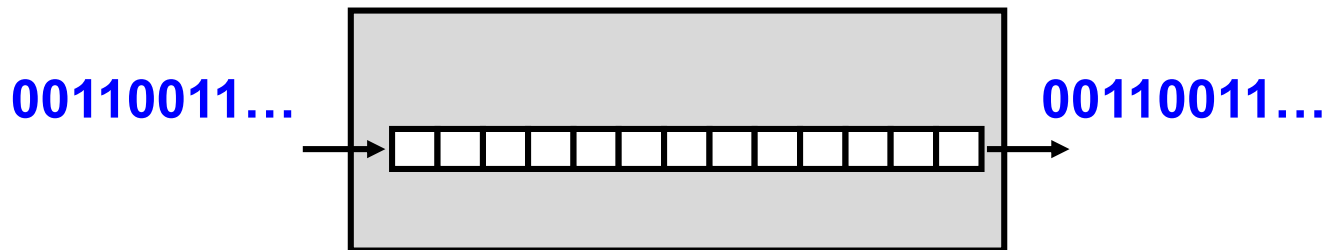
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How to Test FF Scan Chain?

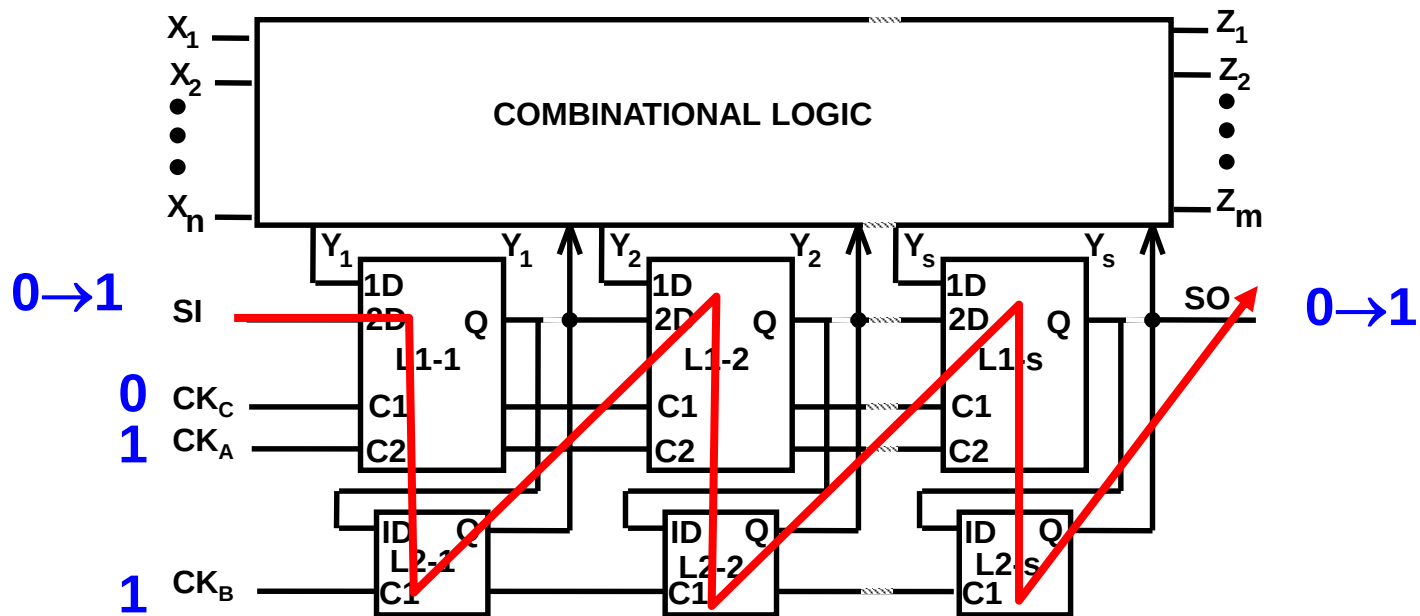
- Scan chains themselves can be faulty. Must be tested too
- Scan chain *integrity test*
 - ♦ A sequence 00110011... is shifted in and shifted out
 - ♦ No capture clock
 - ♦ Each SFF undergo four transitions: 0→1, 0→0, 1→1, 1→0
- Detects all stuck-at faults, transition faults in scan chains



$$\underbrace{(N_{pattern} + 1) \times L}_{\text{shift}} + \underbrace{N_{pattern}}_{\text{capture}} + \underbrace{2L}_{\text{scan integrity}}$$

How to Test LSSD Scan Chain?

- LSSD scan chain **Flush Test**
- $CK_C = 0$, $CK_A = 1$, $CK_B = 1$
 - ♦ A combinational path from SI to SO is formed
 - ♦ Apply **SI = 0 → 1** transition at SI
 - ♦ Measure delay time from SI to SO



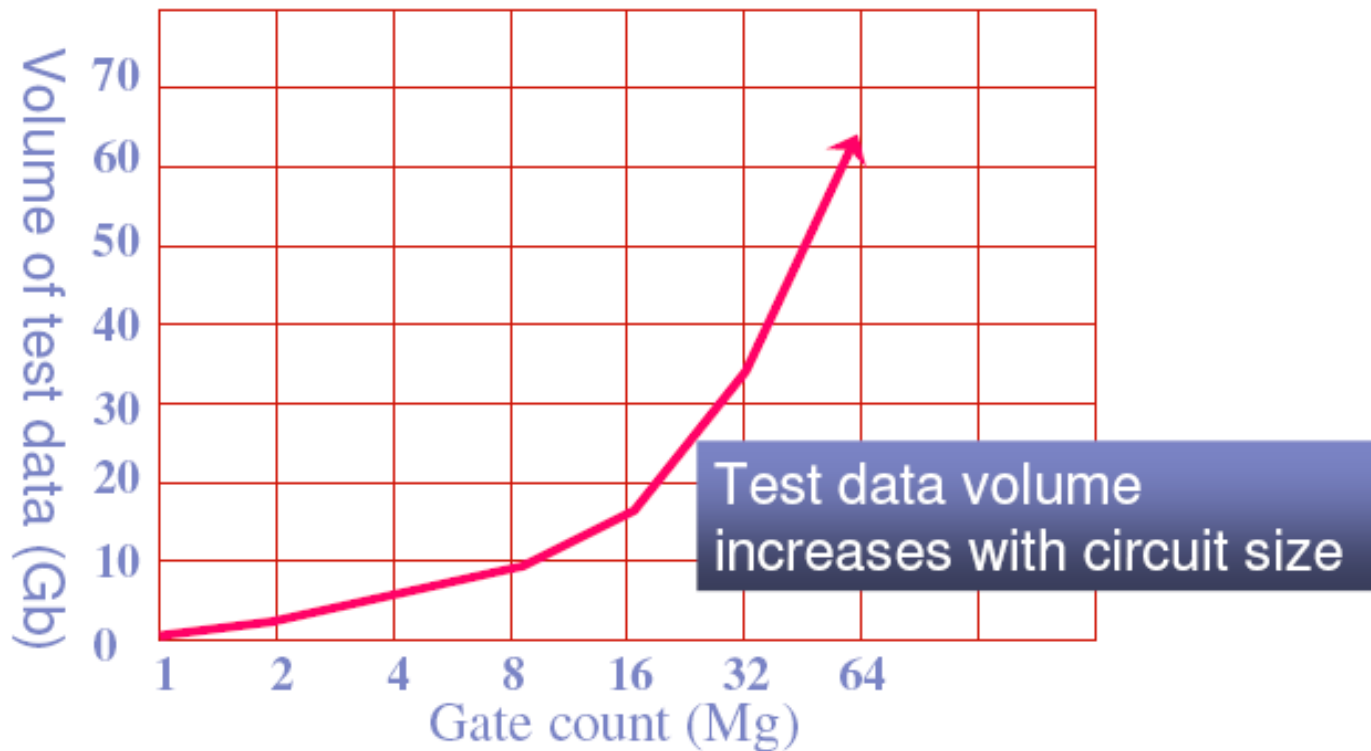
Summary

- Four Issues of scan
 - ◆ Long test time
 - * Multiple scan chains
 - ◆ Large test data volume
 - * see *Test Compression* chapter
 - ◆ Too much overhead
 - * Partial scan
 - ◆ Test scan chain itself
 - * 00110011 or flush test
- Despite above issues, scan is worth doing!

$$(N_{pattern} + 1) \times L + N_{pattern} + 2L$$

FFT

- Q1: We want to reduce test time so we have many scan chains. Can we share scan I/O pins to reduce pin overhead ?
- Q2: Multiple scan chain does NOT reduce test data. What can we do?



(Source: Blyler, Wireless System Design, 2001)

Reference

- [Eichelberger 77] E.B. Eichelberger, and T.W. Williams, "A logic design structure for LSI testability." Proc. of Design Auto. Conf., 1977. (IBM)
- [Funatsu 75] S. Funatsu, N. Wakatsuki, and T. Arima, "Test generation systems in Japan," Proc. Design Auto. Symp., 1975. (NEC)
- [Kobayashi 68] A. Kobayashi, S Matsue, H Shiba "Flip-flop circuit with FLT capability," Proc. of IECEO conf., 1968. (NEC)
- [McCluskey 86] E. J. McCluskey , *Logic Design Principles*, 1986
- [Williams 73] M.J.Y. Williams, J. B. Angell, "Enhancing Testability of Large-Scale Integrated Circuits via Test Points and Additional Logic," IEEE Trans. on Comput., Vol. C-22, Issue: 1, 1973. (Stanford)